

ELECTRONIC MARKETS AND HIGH FREQUENCY TRADING

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- ✓ Is my own and personal opinion
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A bit about myself

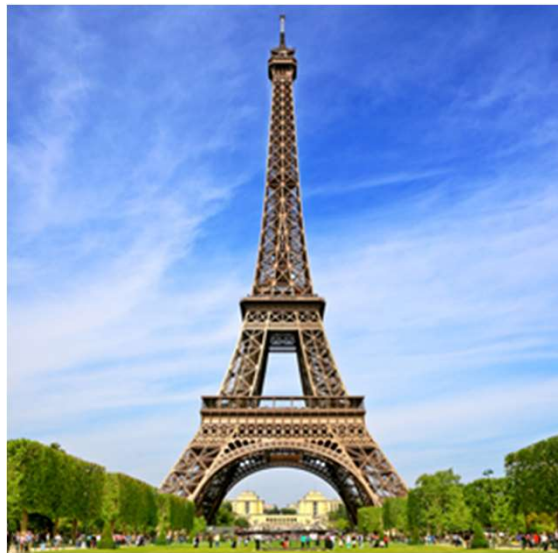


Table of Contents

1. Electronic Markets

- Limit Order Book
- Market orders
- Limit orders

2. High Frequency Trading (HFT)

- Evolution towards electronic markets
- Definition
- Different types of HFT

3. Algorithmic trading

- Simple algorithms: Iceberg, Stop Loss, Take Profit
- Complex algorithms: TWAP, VWAP, PoV
- Comparing complex algorithms

4. Conclusions

1. Electronic Markets

NYSE specialists

- Designated market makers: specialists (NYSE)
 - They are dedicated market makers on a particular security
 - They are obliged to quote buy and ask prices all the time
 - Based on inventory and risk appetite, they change their quotes
 - ❑ Positive inventory → need to sell → aggressive ask prices
 - ❑ Negative inventory → need to buy → aggressive bid prices
 - ❑ Increased risk (market or liquidity) → increased spread



newsok.com

Self-organised liquidity

- In electronic markets, there are **several liquidity providers**:
 - Traders providing liquidity in the market
 - Limit Orders in the Limit Order Book
 - And **several liquidity consumers**:
 - Traders consuming the liquidity of the market
 - Market Orders
 - Traders can change from providers to consumers and vice-versa
 - **No dedicated dealers** or specialists
-
- How can electronic markets operate without dedicated market makers?
 - It is the **aggregate of liquidity providers** that makes the market
 - But providers (generally) have **no obligation** to place both ask and bid quotes
 - **Self-organised liquidity**
 - ❑ Any trader can choose, at any instant of time, to either provide or consume liquidity



geek-whisperers.com

Limit Order Book (LOB)

- The LOB is the **current liquidity of the market** at every available price level
 - It is the aggregate of all Limit Orders (LOs)
- A LOB has 2 sides:
 - **Bid side:** Prices at which the market buys from clients
 - **Ask side:** Prices at which the market sells to clients
- In electronic markets, **dealers are anonymous and real-time data is aggregated:**
 - All you see are the total volumes at every price
 - But you do not know how many shares each trader inserted in the LOB
 - For regulation, stock markets have the ID of every trader and every order

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Market Quality Statistics 

SPDR S&P 500 ETF TR TR UNIT

Orders Accepted 534,996 Total Volume 4,951,439

TOP OF BOOK			LAST 10 TRADES		
	SHARES	PRICE	TIME	PRICE	SHARES
ASKS ↑	14,100	178.7500	10:19:21	178.7100	100
	12,700	178.7400	10:19:21	178.7100	300
	20,000	178.7300	10:19:21	178.7100	200
	11,500	178.7200	10:19:21	178.7100	200
	2,200	178.7100	10:19:21	178.7100	200
BIDS ↓	3,800	178.7000	10:19:21	178.7100	100
	21,200	178.6900	10:19:21	178.7100	100
	16,300	178.6800	10:19:21	178.7050	100
	16,200	178.6700	10:19:21	178.7050	100
	16,900	178.6600	10:19:21	178.7050	100

Last updated 10:19:24

Microstructure of the LOB

- Most common priority is **price/time priority**:
 - First, price priority: LOs at the best price are selected first
 - Second, time priority: Within the best price level, the oldest LOs are selected first
- But there is also **pro-rata priority**:
 - First, price priority as before
 - Second, volume priority: trades are distributed in proportion of the volumes of the LOs
- **Tick**:
 - The smallest price increment in the LO
 - Real prices are discrete, not continuous
- **Spread**:
 - Difference between the best ask price and the best bid price
 - Price of liquidity: the expected gain of a dealer
 - Measure of liquidity:
 - ❑ Highly liquid markets → spread = 1 tick

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BATS SPY 

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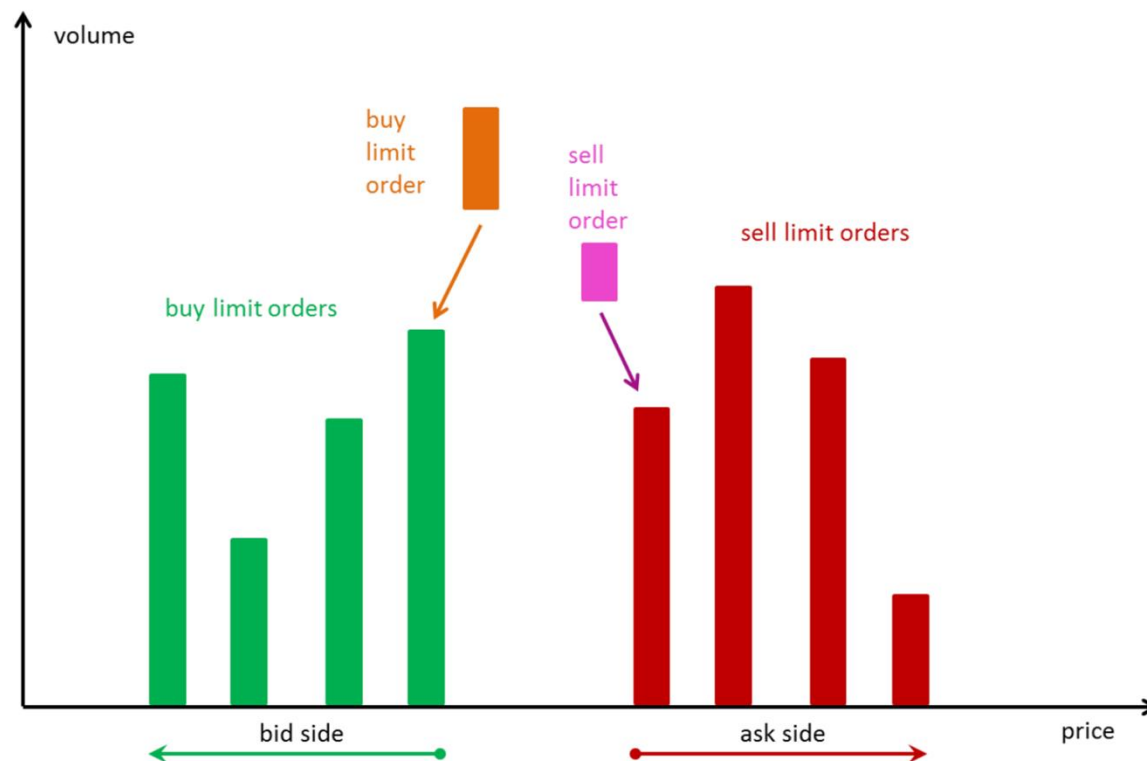
Limit orders vs market orders

- When you send a **buy market order**:
 - You buy at the best selling market price, i.e. the lowest available
 - Your order hits the ask side: buy MO ↔ sell LO
- If you trade via **Limit Orders**:
 - You have a better price: you earn the spread
 - But delayed and uncertain execution:
 - ❑ Your LO sits passively on the LOB, waiting to be matched against an MO
 - Market risk and execution risk
- If you trade via **Market Orders**:
 - You have a worse price: you pay the spread
 - But immediate and certain execution:
 - ❑ Your MO hits the LOB and is matched vs the LOs at the best price
 - No market risk and no execution risk

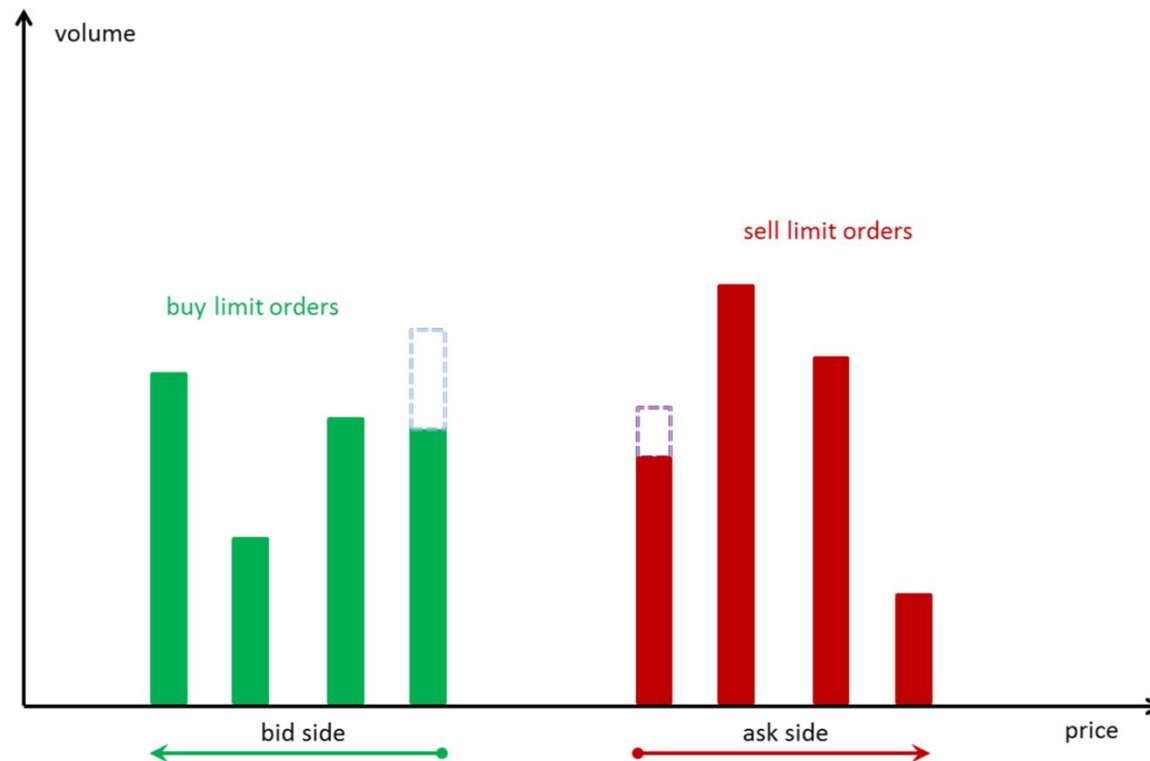


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LOB and LOs



LOB and MOs



2. High Frequency Trading

Evolutions in financial markets

- **Automatisation**

- All markets are essentially electronic
- “The pit” is essentially in the past (except NYSE)

- **Fragmentation**

- Before: Only the primary market existed
- Now: Dozens of different venues where to trade
 - ❑ Primary market, MTFs, dark pools, etc

- **Information**

- Faster, richer real-time data
- Before: the problem was to find data for trading
- Now: the problem is how to filter meaningful data from noise (data mining)



benkepple.com



wired.co.uk

The rise of Algorithmic Trading

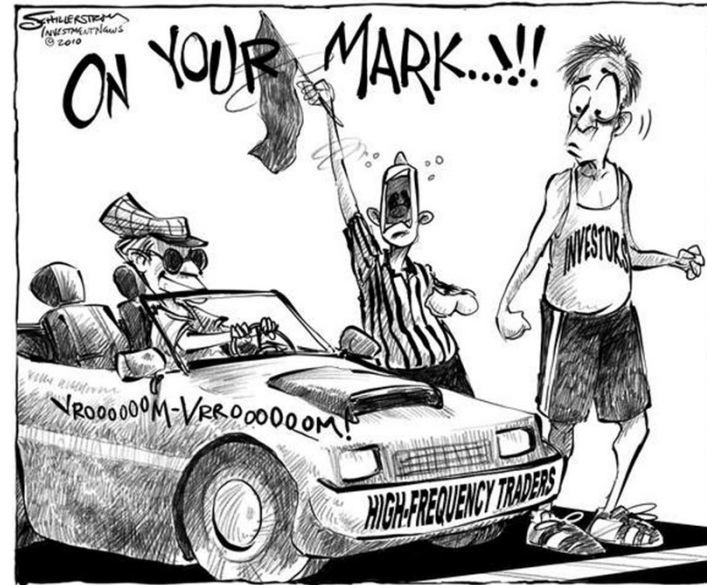
- What is **Algorithmic Trading**?
 - Systematic investment decisions
 - More productivity due to automated processes
 - Widespread use of quantitative tools
- In a sentence: “Algorithms execute whilst humans monitor”



phys.org

The birth of High Frequency Trading

- Algorithmic Trading vs human traders
 - Faster
 - More reactive
 - More accurate
- Information access at very high speeds
 - Low-latency real-time
 - Direct market access
 - Co-location
- Technological advances in computing
 - Data processing
 - Parallel computing
 - Overclocking
- In a sentence: “HFT” are algorithmic traders using speed as their main advantage
- Remark: Ironically, HFT does not necessarily mean very frequent trading !



investmentnews.com

Definition of high frequency trading

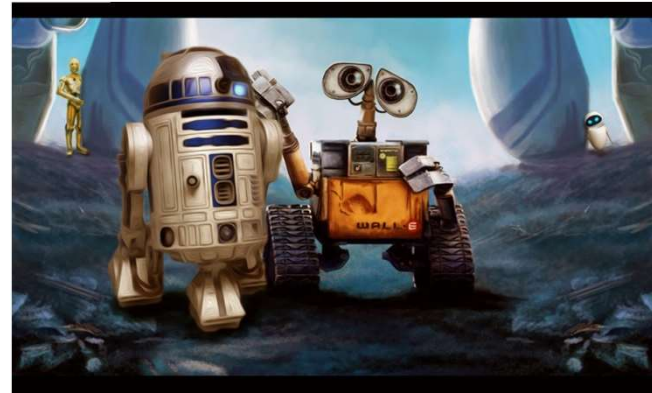
- **Algorithmic trading** firms
 - Using computer algorithms to trade
 - Which automatically determine individual parameters of the order
 - ❑ Time, price, size
 - With limited or no human intervention
 - ❑ An automated system makes decisions on initiation, routing and execution of the order
- **High frequency trading**
 - Subset of algorithmic trading
 - Infrastructure intended to minimise latency
 - No human intervention in order initiation, routing or execution
 - High message intraday rate (orders, quotes, cancellations)
 - ❑ Average of 2 messages per second in a single instrument
 - ❑ Or 4 messages per second in all instruments
- Sources: www.acacomplianceeurope.com & MiFID II



shutterstock.com

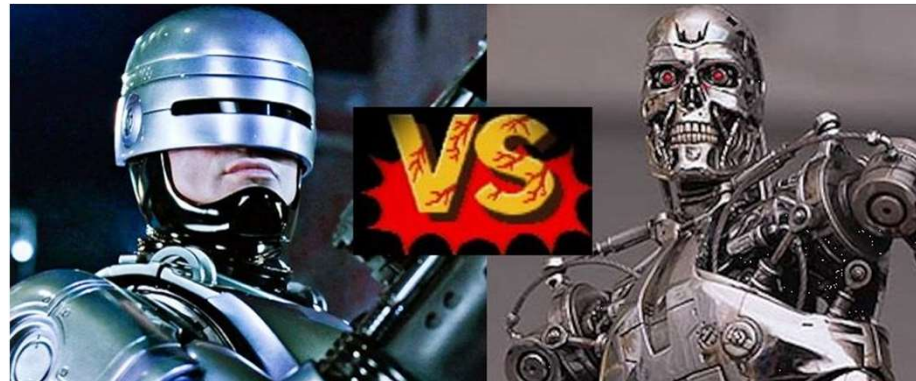
HFTs are not homogeneous

- 4 types of HFTs, according to the SEC:
 - HF Market Making
 - HF Arbitrage
 - HF directional
 - Manipulators



lebeauleblog.files.wordpress.com

- Alternative definition (mine):
 - Makers
 - Takers
 - Gamers



supercoolrobots.com

Makers

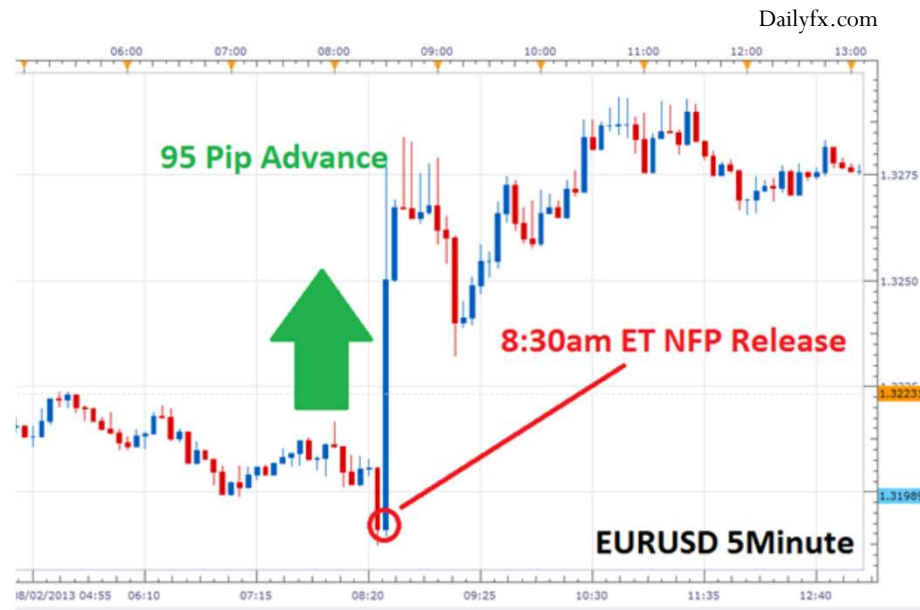
- **Liquidity providers:**
 - Place limit orders in both sides of the Limit Order Book (LOB)
- **Dealers:**
 - They earn the spread by buying at the bid and selling at the ask
- **Speed:**
 - Fast and frequent trades
 - Earn the spread as many times as possible
- **Priority in the LOB:**
 - Be the first in the queue (price and time) for higher probability of execution
- **High reactivity**
 - Quick response to liquidity shocks, market conditions and inventory risk

williambanzai7.blogspot.mx



Takers

- **Liquidity consumers:**
 - They take the liquidity offered by market makers
- They use speed to **seize profitable opportunities** before others:
 - Directional
 - Non directional
- Examples:
 - Macro events
 - News trading
 - Short term alpha
 - Arbitrage



Gamers

- They **exploit structural deficiencies** in electronic markets
- They can play as makers, takers or both
- Example 1: **Spoofing**
 - Place big order, generating a false pressure in the market to push it
- Example 2: **Momentum Ignition**
 - Lure traders to trade in a direction by trading another highly correlated security
- Example 3: **Flash trades**
 - Traders can pay some markets to see incoming orders before they hit the book
- Example 4: **Stuffing**
 - Rapidly send and cancel orders to confuse markets and saturate data lines



Dailyfx.com

3. Algorithmic Trading

Iceberg

- **Goal:** Minimise the market impact of a fully visible big limit order
- An iceberg order of (say) 1000 shares consists of two parts:
 - A **visible part** of (say) 100 shares
 - A **hidden part** of (say) 900 shares
- **Mechanism:**
 - Place the visible part in the LOB at the current best price
 - ❑ Send a first clip of 100 shares
 - When the visible part is executed:
 - ❑ Send a second clip of 100 shares at the same price
 - When the second clip is executed:
 - ❑ Send a third LO of 100 shares at the same price
 - We repeat the iceberg game until full execution
 - ❑ In our example, until the 10th clip of 100 shares is executed
- **Risk:** Incomplete execution if the price moves away from us



www.express.co.uk

Stop Loss & Take Profit

- **Stop Loss:**

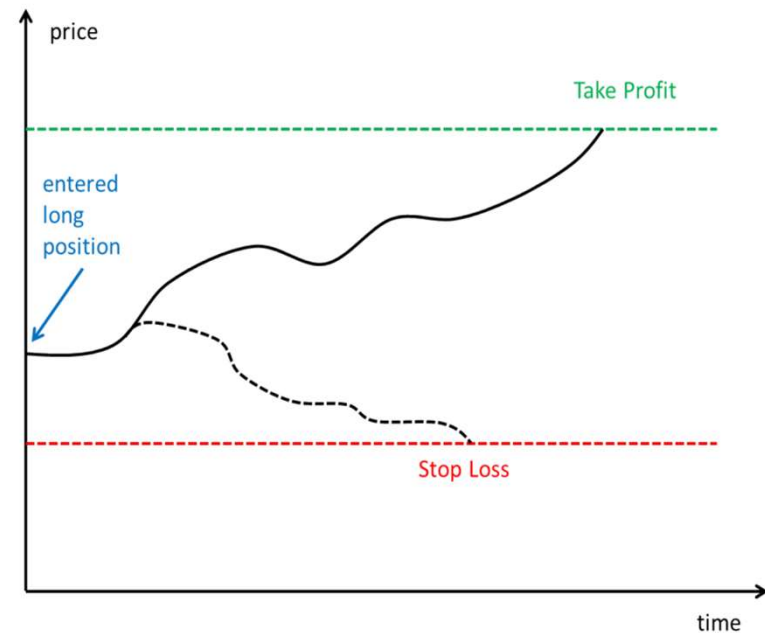
- Set a maximal loss for our portfolio
 - ❑ Set a minimal price or lower bound for our portfolio
- When the portfolio value hits the threshold
 - Get out of the position: sell everything via MO

- **Take Profit:**

- Set a maximal gain for our portfolio
 - ❑ Set a maximal price or upper bound for our portfolio
- When the portfolio value hits the threshold
 - ❑ Get out of the position: sell everything via LO
- The LO can be placed in advance e.g. at entry

- How to determine the thresholds?

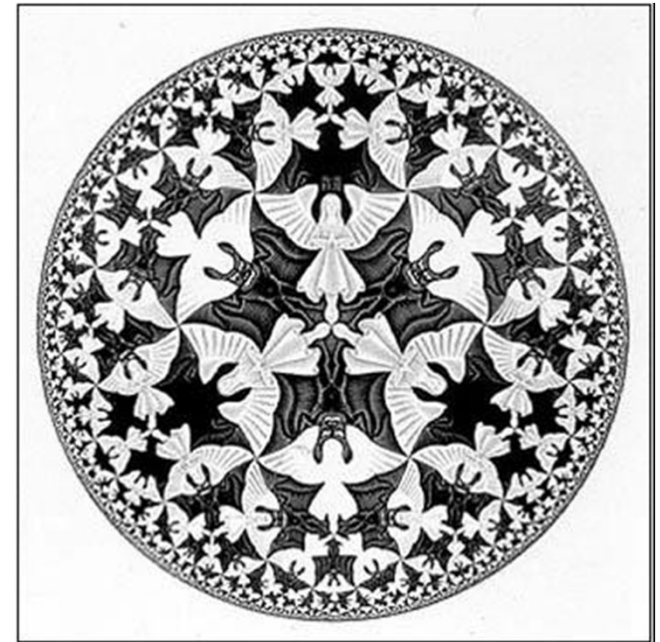
- Investor's risk appetite and expected return
- Volatility of the asset



Time-weighted average price (TWAP)

- **Goal:** Minimise the market impact by spreading a big trade along the day
- **Benchmark:** Average price of the day
 - Example: Average 1-minute mid price
- **Mechanism:** Assume we want to buy **1,000,000 shares**
 - TWAP splits the full trade evenly in time
 - ❑ During 8.5 hours: from 8:00 to 16:30 (London)
 - ❑ $1,000,000 / 8.5 \sim 117,000$ shares per hour
 - ❑ $1,000,000 / (8.5 * 60) \sim \mathbf{1,960 \text{ shares per minute}}$
 - It is a static algorithm:
 - ❑ The schedule is known a priori
 - ❑ The times and quantities of the “child orders” are constant
- **Risks:**
 - Very predictable pattern
 - ❑ Can easily be spotted by HFTs and be “gamed”
 - ❑ We can add some randomness in the number of shares and the times of the child orders

M.C. Escher



Volume-weighted average price (VWAP)

- **Goal:** Minimise the market impact by spreading a big trade along the day
- **Benchmark:** Average price of the day weighted by volume
 - $Market\ VWAP = \frac{\sum P_t V_t}{\sum V_t}$
- **Mechanism:**
 - VWAP splits the full trade based on the **intraday historical volume curve**
 - ❑ High historical volume → bigger child order than TWAP
 - ❖ EU open, EU close, US open
 - ❑ Low historical volume → smaller child order than TWAP
 - It is a static algorithm:
 - ❑ The schedule is known a priori
 - ❑ But sizes are not constant as in TWAP
- **Risks:**
 - Still predictable pattern (but less than TWAP)
 - The historical volume curve can diverge from the real volume curve
 - ❑ Volume shocks, news, etc

Katsushika Hokusai



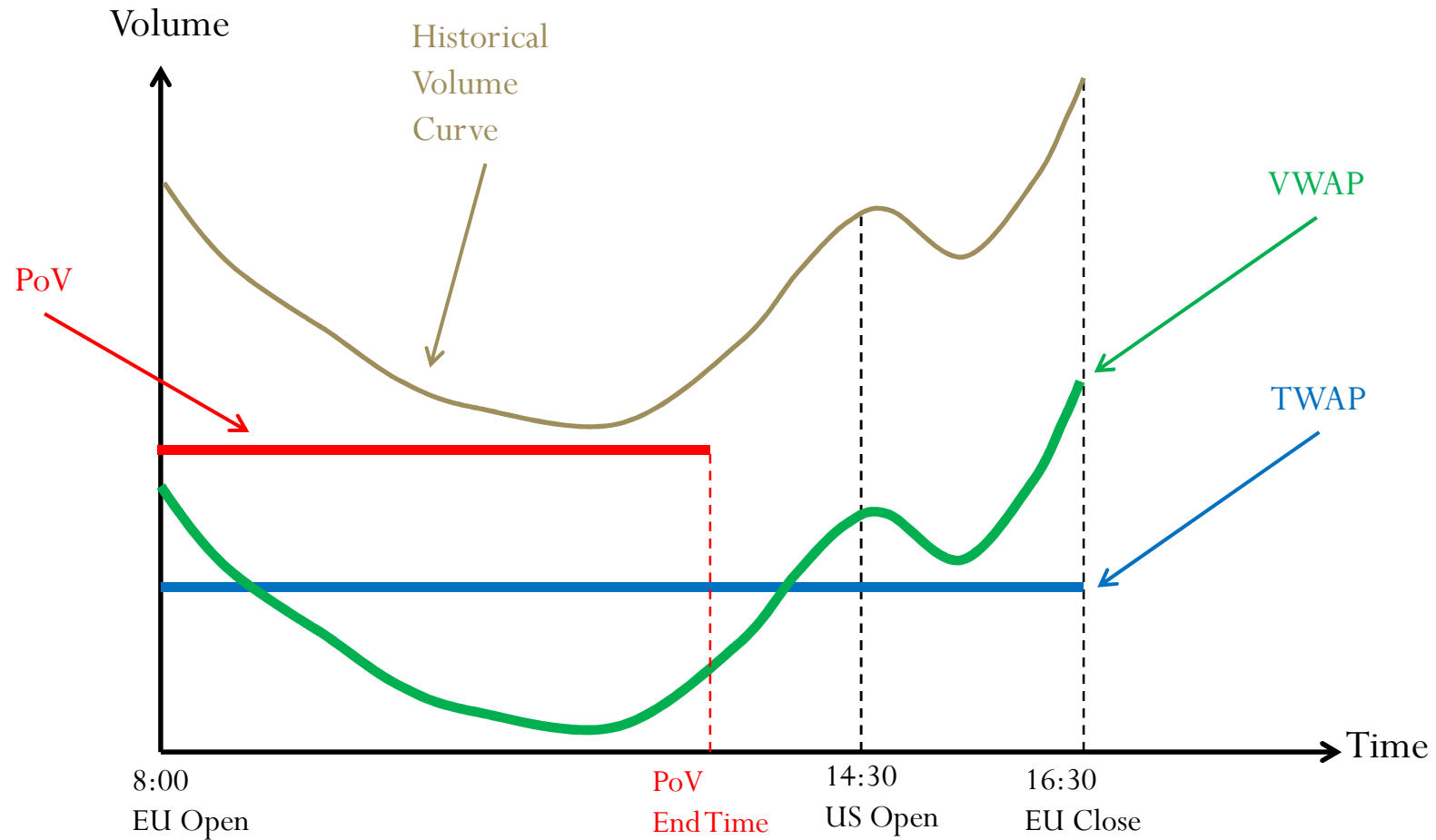
Percentage of Volume (PoV)

- **Goal:** Execute a big order based on a given fraction of the traded volume in the market
- **Benchmark:** Market VWAP of the execution interval
- **Mechanism:** Assume we want to buy **1,000,000 shares**
 - PoV aims a constant participation of (say) 20%
 - ❑ 10,000 shares traded in the market → PoV executed 2,000 shares
 - ❑ 100,000 shares traded in the market → PoV executed 20,000 shares
 - It is a dynamic algorithm:
 - ❑ The schedule and the end time are not known a priori
 - ❑ The algorithm depends on the real volume curve, not the historical volume curve
- **Risk:**
 - Very sensitive to volume shocks
 - We can add some price protection when there are predictable volume shocks
 - ❑ Quarterly results, weekly data release, etc

Eugene Delacroix



Comparing TWAP, VWAP and PoV



4. Conclusions

Summary and conclusions

- Summary
 - We compared dealer markets with electronic markets
 - We explained the Limit Order Book, limit orders and market orders
 - We defined High Frequency Trading
 - We analysed the different types of HFTs
 - We explained simple algorithms: Iceberg, Stop Loss, Take Profit
 - We compared complex algorithms: TWAP, VWAP, PoV
- Conclusions
 - Financial markets will become more electronic, no less
 - ❑ Regulators favour listed markets over OTC markets
 - HFTs are not homogeneous
 - ❑ Not all kinds are toxic to the markets (gamers)
 - ❑ Some can even stabilise markets (makers)
 - Trading algorithms follow predetermined trading rules and patterns
 - ❑ It is important to understand the features and risks of each algorithm
 - A very sophisticated algorithm can be created with several small algorithms

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Thank you for your
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